

Roll No.

TME-707

**B. TECH. (ME) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

AUTOMOBILE ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define automobile. Give the typical specifications of an automobile.

(CO1 & CO2)

(b) Explain the power train for a front wheel driven automobile with its relative advantages.

(CO1 & CO2)

(c) State the requirements of bodies for various types of vehicles.

(CO1 & CO2)

2. (a) What is main purpose of fuel injection system in CI engine ?

(CO1 & CO2)

(b) Explain the functioning of turbocharger with suitable sketch.

(CO1 & CO2)

P. T. O.

(2)

- (c) Explain the working of ignition system with its own electric generator to provide the required necessary energy for the vehicle system.
(CO1 & CO2)
3. (a) What are the functional requirements of a gearbox ? Explain working of a synchromesh gearbox.
(CO3 & CO4)
- (b) What are the benefits of anti-lock brake system ?
(CO3 & CO4)
- (c) Compare the merits and drawbacks of the electric vehicle and conventional automotive.
(CO3 & CO4)
4. (a) Explain with a simple sketch, working of power brake in an automobile.
(CO4 & CO5)
- (b) Explain with a simple sketch, working of splash lubrication system with its relative merits and demerits.
(CO4 & CO5)
- (c) What is Pneumatic and Hydraulic breaking system in a automobile ?
(CO4 & CO5)
5. (a) What are the different types of tyres used in automobile ? (CO5 & CO6)
- (b) What is gasohol, bio-diesel and bio-ethanol ? State their uses.
(CO5 & CO6)
- (c) Compare bio-diesel with diesel engine in performance, combustion and emission characteristics.
(CO5 & CO6)

Roll No.

TME-708

**B. TECH. (ME) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

MECHATRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

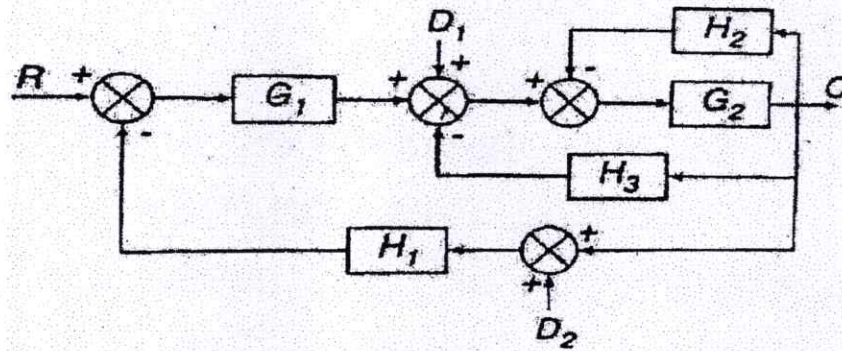
(iv) Each sub-question carries 10 marks.

1. (a) Identify the various system elements for an automatic electronic braking system used in modern automobiles. Design the suitable block diagram for such a mechatronics system by creating the appropriate connectivity between the various constituent elements like electronic control unit, sensors, interface system and the feedback devices. (CO1)
- (b) Explain the *five* performance parameters of a transducer and the points considered while selecting a sensor. (CO3)
- (c) What is the use of Laplace transformation in control system ? Prove that the Laplace transform of . (CO1)

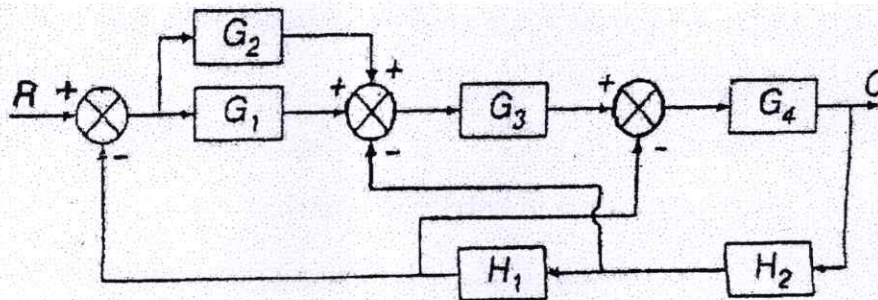
$$e^{-at} \sin \omega t = \omega / (s + \alpha)^2 + \omega^2$$

P. T. O.

2. (a) Derive an expression for the total output for the system represented by the block diagram in below figure : (CO2)



- (b) Derive an expression for the total output for the system represented by the block diagram in below figure : (CO2)



- (c) Represent the following set of equations by a signal flow graph and determine the overall gain relating x_6 and x_1 : (CO2)

$$x_2 = ax_1 + fx_2; \quad x_3 = bx_2 + ex_4$$

$$x_4 = cx_3 + hx_5; \quad x_5 = dx_4 + gx_2$$

3. (a) Explain the different performance terminology which define the performance of transducer. (CO3)
- (b) Explain the *two* types of differential transformer in detail. (CO5)
- (c) Explain the *two* types of optical encoders in detail. (CO5)

(3)

4. (a) Explain Force-voltage and Force-current analogy. (CO4)
(b) For an armature controlled D. C. motor, develop the table for relevant dynamic equations and their corresponding Laplace transforms. (CO4)
(c) For a separately excited D. C. generator connected to load resistance, develop the table of relevant dynamic equations and their corresponding Laplace transforms. (CO4)
5. (a) What are the different types of input signals ? (CO6)
(b) What is the effect of transfer function parameter variations in an open loop and closed loop control system ? (CO6)
(c) Explain the different signal conditioning processes. (CO6)

Roll No.

DEEM-08

B. TECH (ME) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

SUPPLY CHAIN MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the concept of Supply Chain Management ? Explain the different types of Supply Chain Management. (CO1)
- (b) Describe the selection of facility location is helpful in supply chain management. Also write the different methods of selecting facility location. (CO1, CO2)
- (c) Discuss the distribution issues in Supply Chain Management. (CO2)
2. (a) What is logistics ? Explain the Logistics Network configuration is important for supply chain management. (CO2)

P. T. O.

(2)

- (b) Describe the importance of transportation strategy with single service and intermodal services enhance customer satisfaction. (CO1, CO2)
- (c) Explain the objectives of logistics management. Also write the importance of vehicle routing and scheduling helpful for enhancing supply chain management. (CO2, CO3)
- 3. (a) What is inventory management ? Explain the in detail economic order quantity EOQ. (CO2, CO3)
- (b) What is lead time ? How lead time reduction improves the supply chain management system ? (CO4)
- (c) Define the terms, Bullwhip effect, Lead time, Supply Chain integration in detail. (CO4, CO5)
- 4. (a) Describe distribution strategies with distribution requirement planning (DRP). (CO4)
- (b) What is Demand forecasting ? Also write the different methods for forecasting. (CO5, CO6)
- (c) Explain the components of physical distribution systems in SCM. (CO6)
- 5. (a) Describe the main issues of Supply Chain Management (SCM) for national and internationally. (CO5)
- (b) Write the short notes on procurement and outsourcing strategies. What are the five dimensions of SCM ? (CO4, CO5)
- (c) Explain the concepts of supply chain. Also write the role of manager in supply chain management. (CO5, CO6)

Roll No.

DEDE-09

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

FINITE ELEMENT METHOD

Time : Three Hours

Maximum Marks : 100

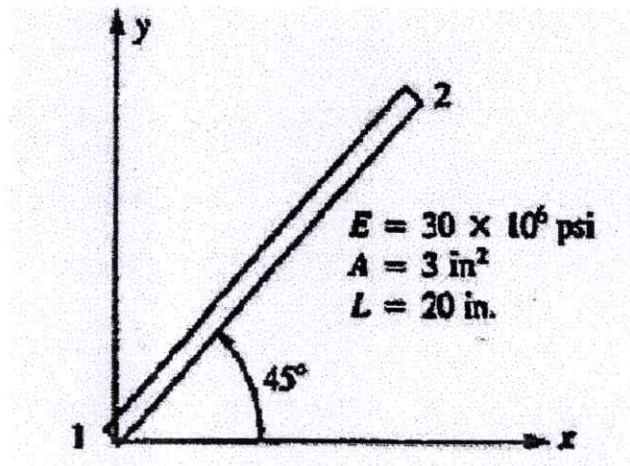
- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) What do you understand by finite element in finite element method ?
(CO1, CO2)
(b) Elaborate the steps for carrying out the finite element procedure.
(CO1, CO2)
(c) What are the difference and similarity between the stiffness matrix of a spring element and bar element ?
(CO1, CO2)
2. (a) Derive the element matrix equation for a bar element. (CO2, CO3)
(b) How do you treat non-homogeneous boundary condition ? (CO2, CO3)
(c) Derive the stiffness matrix for two connected bar elements. (CO2, CO3)

P. T. O.

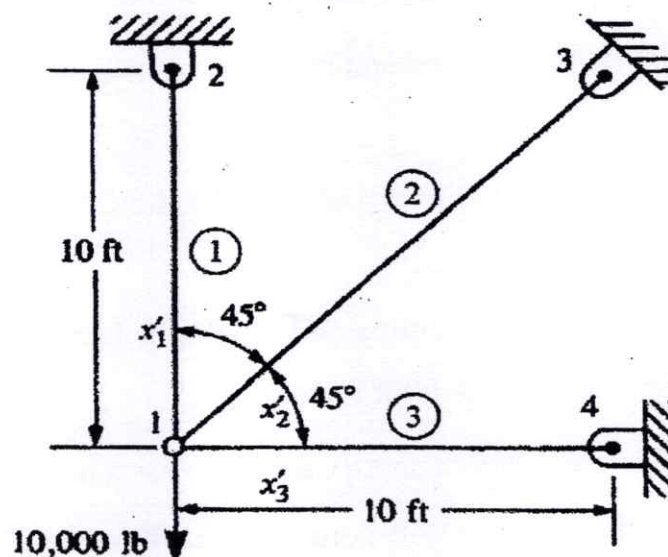
3. (a) Derive the stiffness matrix for a bar element in 2D global coordinated and evaluate it for the bar element shown in the figure below :

(CO2, CO4)



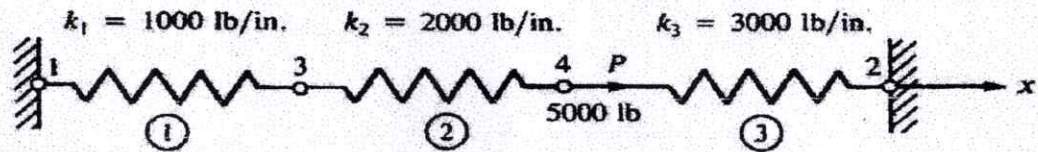
- (b) Calculate the length of element 2, displacement at node 1. $E = 30 \times 10^6 \text{ psi}$, $A = 2 \text{ in}^2$. Node 2 and 3 are at same horizontal level and node 3 and 4 are at same.

(CO2, CO4)



(3)

- (c) Calculate global stiffness matrix, displacements at node 3 and 4 and reactions at node 1 and 2. (CO2, CO4)



4. (a) How many degrees of freedom does a beam element has and why ? (CO4, CO5)
- (b) Derive the element stiffness matrix for beam element. (CO4, CO5)
- (c) How can you combine two beam elements in an assemblage ? Explain with diagram. (CO4, CO5)
5. (a) With diagram, derive the finite element formulation of a spring mass system. (CO5, CO6)
- (b) Derive the finite element formulation of a spring mass system with three equal masses connected in series with springs of same stiffness. (CO5, CO6)
- (c) For spring mass system consisting of two equal masses of 2 kg and connected in series by two springs of stiffness 40 N/m, calculate the two natural frequencies. (CO5, CO6)

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22/12/23

9:30 to 12:30

Roll No.

TAS-701

**B. TECH. (ME/AE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

FLIGHT DYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the steady state performance of an aircraft. (CO1)
(b) Where does lift comes from ? Explain the *three* common explanations. (CO1)
(c) Explain the stagnation point and derive a generalized relation for pitot tube. (CO1)
2. (a) What are the different ways to reduce induced drag ? (CO2)
(b) Define pressure drag and skin friction drag. (CO2)
(c) Define safe range and still air range. (CO2)
3. (a) Define aerodynamic center and center of pressure. (CO3)

P. T. O.

(2)

- (b) With a neat sketch, derive an expression for rate of climb. (CO3)
- (c) Describe briefly the salient features of flap systems. (CO3)
- 4. (a) Derive an expression for power-off glide with a neat sketch. (CO4)
- (b) What do you understand by steep angle glide ? (CO4)
- (c) Derive the power required minimum condition of an aircraft. (CO4)
- 5. (a) How is the drag of a 2D wing different from a 3D wing ? (CO5)
- (b) What is the effect of altitude of maximum and minimum speed of aircraft ? (CO5)
- (c) Derive the drag minimum condition of an aircraft. (CO5)